



Research Paper

More frequent commitments promote cooperation, ratcheting does not

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ABSTRACT

International climate negotiations have so far failed to produce ambitious climate cooperation. We combine laboratory experiments with simulations to investigate the performance of two negotiation design features to address this failure: The Paris Agreement's ratchet-up mechanism, which requires countries to gradually increase their ambition, and a new policy proposal to negotiate more frequently. We find that more frequent interactions allow subjects to build trust and cooperation more safely over time. Conversely, subjects in a ratchet-up design tend to become more cautious to protect themselves from free riders. Thus, more frequent revisions of commitments promote cooperation, but the ratchet-up design fails to achieve the same result.

1. Introduction

The initial commitments of the Paris Agreement on climate change fell substantially short of achieving the agreement's climate objective to limit global warming to well below 2 °C (Liu and Raftery, 2021; Climate Action Tracker, 2024). To address this challenge, the Paris Agreement includes a number of mechanisms to increase ambition. One important example is the ratchet-up mechanism that requires countries to submit Nationally Determined Contributions (NDCs) every five years, with each new NDC being more ambitious than the previous one (UNFCCC, 2015, Article 4). However, in line with results in cooperation and behavioral research (Barrett and Dannenberg, 2016; MacKay et al., 2015; Schmidt and Ockenfels, 2021), the gap between the NDCs and the emissions reductions needed to meet the long-term goals remains substantial, so other proposals are being made to adapt and supplement the agreement. One recent proposal is to increase the frequency of interactions in climate negotiations (Carattini and Löschel, 2021; Reuters, 2021). For example, ahead of the 2025 Conference of the Parties to the UNFCCC, the European Parliament recommended that parties update and strengthen their NDCs more frequently, i.e., every two years instead of every five (European Parliament, 2025, p. 8). However,

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little is known about the effectiveness of such climate negotiation design features. While they appear plausible at first glance, they may actually backfire, for instance if a ratchet-up mechanism leads countries to be more reluctant to commit to climate policies in the first place. Unfortunately, controlled evidence on counterfactual institutional changes in international climate policy is lacking. Therefore, in this study, we investigate the effectiveness of these two important negotiation design features in the context of the Paris Agreement by combining a simulation approach with causal evidence from a highly controlled laboratory setting.

Specifically, we investigate how the ratchet-up mechanism interacts with more frequent decisions in a canonical repeated cooperation game (Isaac and Walker, 1988; Ledyard, 1995). To test these two important design features, both independently and in interaction, we use a full factorial design that varies in two dimensions: Whether contributions must ratchet-up and the frequency with which contributions must be made. The ratcheting condition requires that each contribution to the public good must be at least as large as the contribution in the previous round. The frequency of interactions concerns whether participants with a given endowment contribute to the public good either in one or multiple decisions per round. As a point of departure for guiding intuition and building hypotheses, we use simulations that examine the interaction between the negotiation design features and robust behavioral patterns previously observed in empirical cooperation research – in particular, the willingness of some participants to conditionally cooperate (Fischbacher et al., 2001). This allows us to gain insight into potential behavioral responses to the negotiation design features.

Our study contributes to an experimental literature that examines the impact of behavioral complexities and the institutional context on the level of cooperation in climate negotiations, such as the role of equity preferences (Dannenberg et al., 2010), probabilistic and ambiguous thresholds (Dannenberg et al., 2015), inequality and communication (Tavoni et al., 2011), and different negotiation foci (Schmidt and Ockenfels, 2021). We go beyond this literature by examining the role of incremental and irreversible contributions in the context of the climate dilemma. Schelling (1960) hypothesized that small, sequential, and contingent commitments can foster cooperation in negotiations. A subsequent theoretical literature on Schelling's hypothesis shows that more frequent interactions can indeed increase the provision of public goods, especially when incremental contributions are irreversible (Admati and Perry, 1991; Choi et al., 2008; Gale, 2001; Lockwood and Thomas, 2002; Marx and Matthews, 2000; Simon and Stinchcombe, 1989). Previous experimental studies have tended to confirm this prediction (Dorsey, 1992) and show that participants are particularly willing to cooperate when they can make many contribution decisions and can rapidly respond to the behavior of others (Duffy et al., 2007; Friedman and Oprea, 2012; Kurzban et al., 2001; Oprea et al., 2014). Also, two recent studies have shown that a ratchet-up mechanism may fail to increase cooperation in certain settings (Alt et al., 2023; Gallier and Sturm, 2021). Our simulations and laboratory data bring these different lines of research together, providing robust evidence for the benefits of more frequent commitments in laboratory climate bargaining settings with and without ratcheting, and allowing us to ground the very different interactions of institutional design and behavior in our treatments in the robust conditional cooperation pattern that is ubiquitous in cooperation games.

More specifically, our laboratory experiments show that while the ratchet-up mechanism is ineffective in promoting cooperation, more frequent commitments always lead to substantially higher cooperation. The simulations suggest an explanation, which the data confirm: With ratcheting, participants become more cautious, in order to protect themselves from becoming more vulnerable to exploitation in the future. More frequent interactions, on the other hand, reduce the vulnerability to exploitation, allowing participants to build trust and cooperation more safely over time.

2. Experimental design and procedure

Our laboratory experiment builds on a standard repeated public goods game. At the beginning of the experiment, participants are matched to groups of four, that stay constant for all five rounds. In each round of the experiment $t \in \{1, \dots, T\}$, we endow participant $i \in \{1, \dots, 4\}$ with 100 Experimental Currency Units (ECUs) ($w = 100$; 60 ECUs = 1 Euro). Participants then make $d \in \{1, \dots, D\}$ contribution decisions within each round of the experiment. Each decision requires participants to decide how much of their endowment they want to contribute to the public good ($g_{i,t,d}$). Each member of the group receives a benefit of 0.5 ECU for each ECU contributed to the public good. At the end of each round, i 's cumulative contribution to the public good is given by $g_{i,t} = \sum_{d=1}^D g_{i,t,d}$ and the public good provision level amounts to $G_t = \sum_{i=1}^4 g_{i,t}$. The payoff of participant i is given by:

$$\pi_{i,t} = w - g_{i,t} + 0.5 * G_t.$$

For fully rational and purely self-interested participants, this is a social dilemma. Individual payoff maximization predicts $g_{i,t}^* = 0$ while the maximization of the group's payoff demands $g_{i,t}^{opt} = w$.

We implement four treatments in a 2×2 factorial design. We compare between two settings: A standard setting in which participants play a public goods game with five rounds in which they make only one contribution decision per round. We modify the standard setting, to study the effect of increasing the frequency of interactions. In the modified setting, participants also play a public goods game for five rounds but make five contribution decisions within each round. We implement both settings under two different contribution mechanisms: the standard voluntary contribution mechanism and the ratchet-up mechanism. In the voluntary contribution mechanism, participants' contributions in each round are only constrained by their endowment. In contrast, the ratchet-up mechanism imposes an additional constraint, requiring participants' contributions to weakly increase over rounds. In this way, the

ratchet-up mechanism captures the principle of progressively strengthening climate action under the Paris Agreement, which requires countries' newly submitted NDCs to be more ambitious than their previous commitments. Our experiment builds on the experimental designs of Gallier and Sturm (2021) and Alt et al. (2023), who study the effect of the ratchet-up mechanism in a setting in which participants can only make one contribution decision per round.¹ We go beyond these experiments by investigating how the frequency of contribution decisions per round interacts with the ratcheting mechanism.

In our baseline treatment (BASE 5×1), participants play a standard public goods game for five rounds ($T = 5$) and make only one contribution decision ($D = 1$) in each round. Participants play a voluntary contribution mechanism and decide how much of their endowment they want to contribute to the public good. In each round, participants' contributions are constrained only by their endowment and can range from zero to their endowment, i.e., $0 \leq g_{i,t} \leq w$. We have three more treatments: BASE 5×5 , RAT 5×1 , and RAT 5×5 .

In BASE 5×5 , participants also play a repeated standard public goods game with five rounds ($T = 5$). Unlike in BASE 5×1 , participants in BASE 5×5 make five contribution decisions ($D = 5$) within each round. In each decision, participants choose how much to contribute to the public good and their cumulated contributions per round are constrained by their endowment. In the first decision ($d = 1$), their contribution can range from zero to their initial endowment, i.e., $0 \leq g_{i,t,1} \leq w$. From the second decision onwards ($d > 1$), participants' contribution can range between zero and their disposable endowment, i.e., the proportion of their endowment that they have not contributed in one of the previous decisions: $0 \leq g_{i,t,d} \leq w - \sum_{d=1}^{d-1} g_{i,t,d}$.

Participants in RAT 5×1 play a repeated public goods game with five rounds ($T = 5$) and make only one contribution decision ($D = 1$) in each round. RAT 5×1 builds on BASE 5×1 , but applies the ratchet-up mechanism, which requires participants to weakly increase their contributions to the public good across rounds. More precisely, each contribution per round must be at least as high as in the previous round. In the first round ($t = 1$), participants' contributions can range from zero to their endowment, i.e., $0 \leq g_{i,t} \leq w$. From the second round onwards ($t > 1$), contributions are additionally constrained from below by the contributions in the previous round, i.e., $g_{i,t-1} \leq g_{i,t} \leq w$.

In RAT 5×5 , participants play a repeated public goods game with five rounds ($T = 5$) and make five contribution decisions ($D = 5$) in each round. This treatment builds on BASE 5×5 and by implementing the ratchet-up mechanism, it requires that the cumulated contributions in each round are at least as high as the cumulated contributions in the previous round. In the first round ($t = 1$), participants' cumulated contributions are only constrained by their endowment. In the first decision ($d = 1$), their contributions can range between zero and their initial endowment, i.e., $0 \leq g_{i,t,1} \leq w$. From the second decision onwards ($d > 1$), their contributions are limited by their disposable endowment, i.e., $0 \leq g_{i,t,d} \leq w - \sum_{d=1}^{d-1} g_{i,t,d}$. From the second round onwards ($t > 1$), participants' contributions are constrained by their cumulated contributions in the previous round. In the first decision ($d = 1$) of a new round, participants' contributions must be at least as high as the cumulative contribution in the previous round, i.e., $g_{i,t-1} \leq g_{i,t,1} \leq w$.² From the second decision onwards ($d > 1$), contributions can range between zero and the disposable endowment, i.e., $0 \leq g_{i,t,d} \leq w - \sum_{d=1}^{d-1} g_{i,t,d}$.

The experiment was approved by the Institutional Review Board of the Faculty of Management, Economics and Social Science at the University of Cologne (Appendix E).³ We conducted 20 sessions, including a pilot, online via the Cologne Laboratory of Economic Research (CLER). We recruited students from CLER via ORSEE (Greiner, 2015) and programmed the experiment in oTree (Chen et al., 2016). We followed the protocol by Buso et al. (2021) to conduct our online visually monitored sessions and used a widely used web conferencing platform to moderate and monitor the sessions. The protocol guaranteed that the experimenter could talk publicly to all participants. While participants could communicate privately with the experimenter, they could not see or communicate with each other. At the beginning of each session, we informed participants about the general procedure. Participants then read on-screen instructions that explained the rules of the game (Supplementary Material). During the experiment, participants were fully informed. After each decision and at the end of each round, they were given information about the individual contributions of all group members and the corresponding payoffs. If participants had questions, they could use a chat function implemented in the web conferencing platform. Sessions lasted approximately 60 min and participants were paid the sum of the payoffs per round. The earnings ranged from 5.25 Euro to 18.60 Euro, with an average of 13.10 Euro, including a show-up fee of 1 Euro. We paid participants via bank transfer.

A total of 368 participants took part in our experiment, all gave informed consent. By having 22 groups of four participants each for BASE 5×1 and RAT 5×1 and 24 groups each for BASE 5×5 and RAT 5×5 , we reached a total of 92 independent observations. This allows us to detect frequency effects comparable to our prior at a conventional level of statistical significance (5%) at a very high level of statistical power (Appendix A).

¹ In contrast to the previous studies by Gallier and Sturm (2021) and Alt et al. (2023), our experiment relies on a linear public goods game instead of a non-linear version of the game.

² Different rules could have been applied in RAT 5×5 to ensure that the cumulative contributions in a given round of the game were at least as high as the cumulative contributions in the previous round. For example, an alternative rule could allow the first decision to be unconstrained, while specifying that any potential shortfall is automatically compensated in the last decision of each round. We believe that our rule is easy to communicate and has the most immediate consequences. We leave it to future research to investigate whether the effect of the ratchet-up mechanism in the 5×5 -setting depends on the specific decision in which the condition is enforced.

³ At the time of IRB approval, the experimental design and procedure were fully specified, but we did not pre-register our hypotheses and tests. Our null hypotheses posit no effect of the two negotiation design features, and we apply straightforward two-sided statistical tests throughout. To further guide intuition and to develop directional expectations, we draw on simulations anchored in established patterns of conditional cooperative behavior (Fischbacher et al., 2001), complemented by previous evidence on the ratchet-up mechanism (Gallier and Sturm, 2021; Alt et al., 2023) and on the effects of more frequent interactions (Schelling, 1960).

3. Simulation studies to guide intuition

We use a simulation approach to guide intuition and generate potential directional predictions about how the ratchet-up mechanism and more frequent interactions may affect the provision of a public good. We apply simulations because the predictive and explanatory power of the standard theoretical model, which assumes fully rational and purely self-interested players, is limited in this context. While by the standard backward induction argument, rationality and selfish preferences imply no contributions to the public good in any round and in any of our treatments, it is, however, well-established in the literature that cooperation can emerge, based on robust patterns of (conditional) cooperation observed in the laboratory (Chaudhuri, 2011; Fischbacher et al., 2001; Fischbacher and Gächter, 2010; Ledyard, 1995) and the field (Diederich and Göschl, 2018; List, 2011). In the literature, there are generally three robust behavioral patterns in cooperation games: (i) some agents selfishly defect, (ii) others are conditionally cooperative, i.e., willing to cooperate but only as long as others do, and (iii) initial cooperation is very heterogeneous (the heterogeneity can be interpreted as some agents are willing to lead by example and contribute more to the public good than others, especially initially; Eisenkopf and Kölpin, 2023; Gächter et al., 2012). Our simulation approach allows us to study and predict how these conditional cooperative behavioral patterns potentially interact with ratcheting and more frequent interactions in a public goods game.⁴

For our simulations, we randomly sample (with replacement) 1000 groups of size $n = 4$ from a large population of artificial agents. Following the literature on conditional cooperation in social dilemmas (Ambrus and Pathak, 2011; Axelrod, 1984; Fischbacher et al., 2001; Fischbacher and Gächter, 2010; Kreps et al., 1982; Ockenfels, 1999), our simulations focus on the interaction of different types of agents. Our population consists of an equal number of agents who selfishly defect (DFs) and conditional cooperators (CCs). The simulations follow a two-step procedure applied to both the 5×1 - and 5×5 -settings.

In the 5×1 -settings, agent i starts in a first step by choosing her contribution in the first round of the game (Round I): $g_{i,I}$. DFs always contribute zero.⁵ CCs' initial contributions are drawn from a uniform distribution. The lower bound of this distribution is set to zero, while we vary the upper bound to capture different scenarios. In a second step, agent i determines her contribution for the second round (Round II): $g_{i,II}$. CCs' contributions in the second round depend on their group members' contributions in the previous round and their own previous contributions. More precisely, CCs choose a linear combination between the average contributions of the other group members and their own contribution in Round I. In this way, our simulation captures that agents respond differently depending on whether they have been free ridden by their group members who contributed less to the public good than they did or whether they themselves have been free riding on the contributions of their group members. If an agent contributes more than the average of her other group members in Round I, she sets her Round II contribution equal to the average of the other group members' contributions in Round I. If an agent contributes less than her group members in Round I, she sets her Round II contribution equal to the average of her own Round I contribution and the average of her group members' Round I contributions.⁶ Thus, agents who contribute more than their group members reduce their contributions by a larger amount than agents who contribute less than their group members increase their contributions (Blanco et al., 2011; Nunnari and Pozzi, 2022). In RAT 5×1 , an additional restriction applies: Each contribution must be at least as high as in the previous round, i.e., $g_{i,II} \geq g_{i,I}$. This sequential process repeats round by round until the end of the game at the end of Round V.

In the 5×5 -settings, agent i starts with choosing her contribution in the first interaction of the game (Decision 1 in Round I): $g_{i,I,1}$. As in the 5×1 -setting, DFs always contribute zero and CCs' initial contributions are drawn from a uniform distribution with a lower bound of zero and an upper bound that varies by scenario. In a second step, agent i then determines her contribution for the second decision (Decision 2) in Round I. To maintain compatibility with the 5×1 -settings, we apply the same round-to-round process used in the 5×1 -settings to model CCs' contributions across decisions within each round in the 5×5 -settings. If an agent's contribution in Decision 1 of Round I exceeded the average of her group members, she sets her contribution in Decision 2 to the average of the other group members' contributions in Decision 1. If an agent contributes less than her group members on average in Decision 1, she sets her contribution in Decision 2 equal to the average of her own previous contribution and the average of her group members' contributions in the previous decision. This sequential process continues decision by decision throughout all five decisions of Round I, thereby determining agent i 's cumulative contribution in Round I, $g_{i,I} = \sum_{d=1}^5 g_{i,I,d}$. At the beginning of Round II, CCs base their contributions in the first decision, $g_{i,II,1}$, on the cumulative contributions in Round I. Here, $g_{i,II,1}$ is a scenario-specific proportion of the average of the own cumulative contribution in Round I and the average of the others' cumulative contributions if i 's cumulative contribution is below the other's average cumulative contributions in Round I. If i 's cumulative contribution is above the other's average cumulative contributions in Round I, $g_{i,II,1}$ is a proportion of the other's average cumulative contributions in Round I. In RAT 5×5 , the ratchet-up mechanism also applies. The contribution in the first decision of a given round must be at least as high as the cumulative contribution in the previous round, i.e., $g_{i,II,1} \geq g_{i,I}$. This sequential process repeats round by round until the end of the game in Round V. See Appendix B for further details.

Based on these conditional cooperative behavioral patterns, the simulation in Fig. 1a shows our benchmark and predicts a negative

⁴ Our approach follows several papers at the intersection between economic experiments and simulation studies. See, for example, Carpenter (2007); Duffy (2006); Fischbacher and Gächter (2010); Miller and Andreoni (1991); Sethi and Somanathan (2003). For a general discussion of the complementarities between these two methods, see, e.g., Contini et al. (2006).

⁵ Following Fischbacher and Gächter (2010) we keep the behavior of the DFs in our simulations as simple as possible. Ambrus and Pathak (2011) present a more complex framework that demonstrates how even selfish players may be incentivized to contribute to the public good. For instance, in a finitely repeated game, their contributions can elicit positive responses from conditionally cooperative agents.

⁶ For the linear combination in this case, we used the middle of the corresponding interval for simplification reasons.

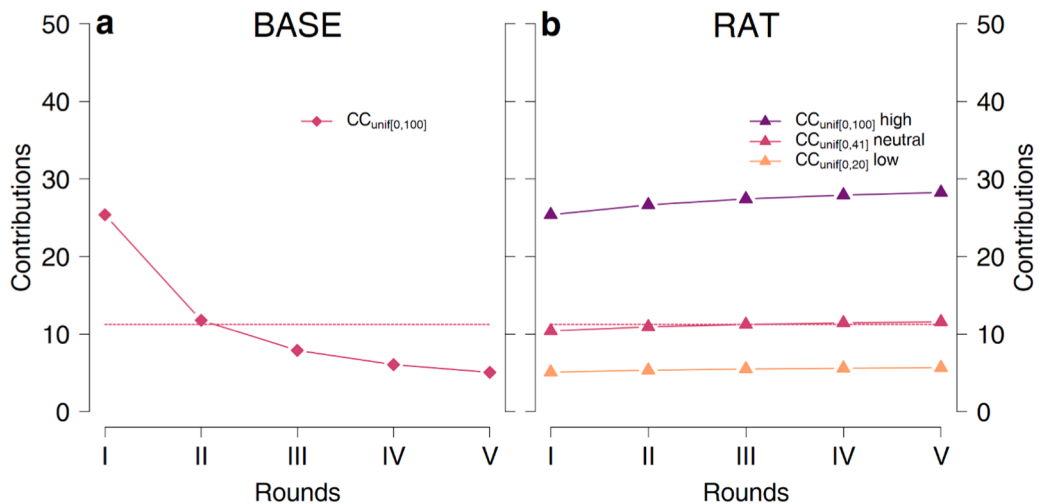


Fig. 1. Predicting contributions in BASE 5×1 and RAT 5×1 .

Note: Simulation results show predicted contributions in ECU per scenario in BASE (a) and RAT (b). $CC_{unif}[0, \{100, 41, 20\}]$: 50% defecting types (DFs) and 50% conditionally cooperative types (CCs). DFs contribute zero. CCs' contributions in Round I are drawn from a uniform distribution in $[0, \{100, 41, 20\}]$. The corresponding predicted contributions in Round I are equal to $0.5 \times 0.5 \times \{100, 41, 20\} = \{25, 10.25, 5\}$. The horizontal dotted lines indicate the average of the predicted contributions for $CC_{unif}[0, 100]$ in BASE 5×1 and $CC_{unif}[0, 41]$ in RAT 5×1 , respectively. See Appendix B for further details.

trend in ECUs contributed to the public good in BASE 5×1 , as indeed has been previously predicted and typically observed in such dilemma games (Fischbacher and Gächter, 2010). CCs reciprocate the behavior of other agents in the previous interaction, and they respond more strongly when being exploited than when being more selfish than others. Over time, this process compresses individual contributions and thus reduces the negative trend.

Fig. 1b summarizes the results of three different scenarios for the ratchet-up mechanism in the 5×1 -setting with low, neutral, and high initial contributions. By design, in contrast to BASE, contributions in RAT cannot decrease over time, and they in fact tend to increase, driven by CCs with contributions below their other group members' contributions. The increase in contributions diminishes over time and eventually reaches a ceiling, because CCs' contributions are increasingly compressed over time and limited due to DFs' defections.

Whether ratcheting increases cooperation depends on initial cooperation rates in Round I. Ex-ante, it is unclear how the initial contributions in RAT 5×1 compare to those in BASE 5×1 . To address this uncertainty, we simulate three different scenarios, shown in Fig. 1b, to predict different potential outcomes. The neutral scenario is chosen such that it yields exactly the same total cooperation as BASE. This is the case when CCs in Round I cooperate approximately slightly less than half of their initial BASE cooperation. However, because ratcheting removes CCs' option to reciprocally decrease their contributions if others free-ride, the same total level of cooperation comes with greater vulnerability against exploitation.⁷ Following the literature on the risk of exploitation in social dilemmas (e.g., De Cremer, 1999; Cubitt et al., 2017), we hypothesize that players initially tend to be more cautious and cooperate much less with ratcheting than in our baseline to mitigate the increased risk of being exploited by free riders. In line with this, Gallier and Sturm (2021) and Alt et al. (2023) report that low initial contributions under the ratchet-up mechanism prevent total contributions from exceeding those in a voluntary contribution mechanism. So, while the precise level of the initial contributions is unknown ex-ante and ratcheting may lead to more or less cooperation (as illustrated by the high or low scenario, respectively), our hypothesis follows the previous empirical evidence and suggests that the limited flexibility to respond to exploitation leads to initial contributions at or below the neutral scenario, undermining the desired effect of the ratchet-up mechanism or may even backfire.⁸

Next, we use our simulations to show how the possibility to make more frequent contributions per decision, i.e., the average of decisions 1 to 5 across all five rounds, in BASE 5

⁷ Agents who contribute to the public good inevitably expose themselves to the risk of being exploited by free riders who benefit from their contributions without contributing themselves. Accordingly, one possible explanation for agents contributing little or nothing to the public good is their attempt to minimize the monetary (e.g., Yamagishi and Sato, 1986; Parks and Hulbert, 1995) and non-monetary (e.g., Aimone et al., 2015; Bohnet et al., 2008) costs of being exploited by free riders. Following Gallier and Sturm (2021), we capture the risk of exploitation in our game as the product of the extent to which agents contribute more to the public good than their group members and the number of times they can be exploited. Ratcheting increases the risk of exploitation because agents can be exploited repeatedly. Since agents cannot reduce their contributions after being exploited, they are unable to negatively reciprocate uncooperative behavior of others.

⁸ Only if the initial contributions in RAT 5×1 exceed the initial contributions in the neutral scenario (10.25), our simulations predict higher overall contributions in RAT 5×1 than in BASE 5×1 (11.2). Fig. 1b also shows that lower initial contributions in Round I lead to a smaller increase in contributions over rounds, because less initial variation limits the scope for increasing contributions by CCs.

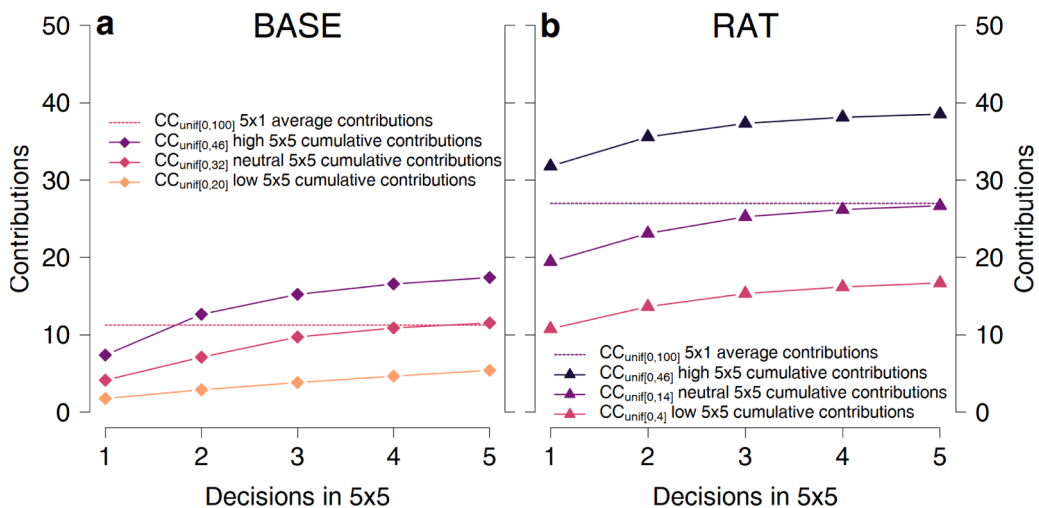


Fig. 2. Predicting contributions in BASE 5×5 and RAT 5×5 .

Note: Simulation results show predicted contributions in ECU per scenario in BASE (a) and RAT (b). For the 5×1 -settings, simulations show the predicted average across all five rounds. For the 5×5 -settings, simulations show the average of the predicted cumulative contributions per decision across all five rounds, e.g., the average for decisions 1 to 5 in all five rounds. $CC_{unif}[0,100]$ 5×1 average of the predicted contributions in 5×1 -settings (see Fig. 1). $CC_{unif}[0,46,32,20,14,4]$ 5×5 predicted cumulative contributions per decision across rounds for CCs' initial contribution in $[0, \{46,32,20,14,4\}]$. See Appendix B for further details.

$\times 5$. It is natural to expect that contributions in the 5×5 -setting also start below those in the first round of the 5×1 -setting (Schelling, 1960). Ex-ante we do not know to what extent contributions in BASE 5×5 start below those in BASE 5×1 . We use our simulations to determine how much initial contributions in BASE 5×5 can fall below those in BASE 5×1 such that the cumulative contributions at the end of the round in BASE 5×5 still exceed the average contributions per round in BASE 5×1 . Fig. 2a summarizes three different scenarios, with low, neutral, and high initial contributions. The neutral scenario is chosen such that CCs' initial contributions in BASE 5×5 lead to cumulative contributions at the end of the round that exactly match the average contributions in BASE 5×1 . More frequent commitments have a positive effect on total contributions, if the initial cooperation per decision in BASE 5×5 is higher than in the neutral scenario, e.g., if the initial cooperation per decision is one fifth of the Round-I-level in BASE 5×1 (high scenario). Following Schelling's hypothesis, we suggest that it is plausible in this treatment to assume that initial contributions start above the neutral scenario, because more frequent interactions allow players to more closely build trust and cooperation over time, thus reducing the vulnerability against exploitation.⁹

Our simulations also allow us to assess the relative performance of the ratchet-up mechanism compared to the opportunity to make more frequent contributions. Comparing Fig. 1 and Fig. 2, we see that the gap between initial contributions in BASE 5×1 and those in the neutral scenario in BASE 5×5 (25 vs. 4.1) is larger than the corresponding gap between the initial contributions in BASE 5×1 and those in the neutral scenario in RAT 5×1 (25 vs. 10.25). Consequently, the range of initial contributions that result in higher contributions than in BASE 5×1 is larger in BASE 5×5 than in RAT 5×1 . This implies that more frequent contributions are more likely to promote cooperation than the ratchet-up mechanism.

Fig. 2b shows the predicted average cumulative contributions per decision in RAT 5×5 . Again, we use three different scenarios (low, neutral, high) for different initial contributions. If the initial cooperation is relatively low, below the neutral scenario, cumulative contributions in RAT 5×5 do not reach the average contributions in RAT 5×1 .¹⁰ If, however, the initial contributions per decision in Round I are relatively large, above the neutral scenario, cumulative contributions in RAT 5×5 exceed the average contributions in RAT 5×1 , for instance, in the high scenario where the initial contributions per decision in Round I are one fifth of the initial RAT 5×1 cooperation. Again, the latter effect appears to be more plausible as more frequent interactions reduce the vulnerability against exploitation. Similar to the 5×1 -setting, the downward momentum of the contributions per decision in BASE 5×5 and the upward momentum of the cumulative contributions per decision in RAT 5×5 tend to decrease over time.

To summarize, our simulations show that the initial contribution levels determine the relative performance of the 5×5 -settings. Since the initial contributions are ex-ante unknown, it remains an empirical question how the possibility to make more frequent

⁹ Only if the initial contributions per decision in BASE 5×5 are higher than those in the neutral scenario (4.1), our simulations predict that the cumulative contributions in BASE 5×5 exceed the average contributions in BASE 5×1 (11.2). In all scenarios, the average cumulative contributions increase, but the increase depends on the initial contributions. The lower the level and thus the variation of the initial contribution, the lower the increase in cumulative contributions.

¹⁰ Only if the initial contributions per decision in RAT 5×5 are higher than those in the neutral scenario (19.5), our simulations predict that the cumulative contributions in RAT 5×5 exceed the average contributions in RAT 5×1 (27.1). As in the 5×1 -setting, a lower contribution in the first decision of Round I is equivalent to less initial variation and leads to a smaller increase in cumulative contributions over decisions.

commitments will affect public good provision levels. That said, our simulations based on conditional cooperative behavioral patterns and in conjunction with previous empirical evidence suggest that ratcheting is less powerful, or even backfires, when promoting cooperation than more frequent interactions. The reason is that analogous behavior to BASE increases the scope of being exploited with ratcheting, but reduces this scope with more frequent interactions.

Finally, we conduct a sensitivity analysis in which we systematically vary the composition of our population to stress-test the robustness of our results (see Appendix B and D). More precisely, we introduce two additional main scenarios: 30–70 and 70–30. In the 30–70 scenario, our population consists of 30% CCs and 70% DFs, whereas in the 70–30 scenario, the composition is reversed – 70% CCs and 30% DFs. The sensitivity analysis reveals that the main results remain robust to changes in the composition of the population and that ratcheting is more likely to promote cooperation compared to BASE when the proportion of DFs in the population is relatively high (see Fig. D1 and D2 for the 5×1 -setting and Fig. D3 and D4 for the 5×5 -setting in Appendix D).¹¹

4. Laboratory results

To estimate the overall effects of our treatment interventions, we aggregate observations across all five rounds of the experiment. Fig. 3a shows, in line with our simulations, that overall public good provision levels, measured in ECU, are significantly higher when there are more frequent interactions, both in BASE and RAT. In BASE, contributions to the public good in the 5×5 -setting (59.5) are almost twice as high as in the 5×1 -setting (31.5, $p < 0.001$; Appendix C, Table C1).¹² Similarly, with ratcheting, contributions to the public good increase from 28.7 in the 5×1 -setting to 58.4 in the 5×5 -setting ($p < 0.001$; Appendix C, Table C1). The ratchet-up mechanism, in contrast, does not promote cooperation. The difference between BASE and RAT is not statistically significantly different from zero, neither in the 5×1 -setting (31.5 vs. 28.7, $p = 0.535$; Appendix C, Table C1) nor in the 5×5 -setting (59.5 vs. 58.4, $p = 0.877$; Appendix C, Table C1).

Next, we extend the analysis by investigating the trends in contributions to the public good over the course of the experiment. Fig. 3b displays groups' average contributions to the public good per treatment, round, and decision. Contributions in BASE 5×1 are very similar to previous results in voluntary contribution experiments (Ledyard, 1995; Fischbacher and Gächter, 2010; Chaudhuri, 2011), both in terms of the average level of the public good provision and the pattern across rounds. On average, participants contribute 31.5 to the public good. Furthermore, contributions decrease from 46.4 in Round I to 11.1 in Round V ($p < 0.001$; Appendix C, Table C2 – Model 1).

Contributions to the public good in Round I of RAT 5×1 are significantly lower than those in BASE 5×1 (20.9 vs. 46.4, $p < 0.001$; Appendix C, Table C2 – Model 1). Even though contributions in RAT 5×1 increase from round to round and are significantly higher than those in BASE 5×1 in Round V (11.1 vs. 35.6, $p < 0.001$; Appendix C, Table C2 – Model 1), the opposing trends are not strong enough to compensate for the initial loss of cooperation due to more cautious initial contributions.

Contributions in the 5×5 -settings show very similar patterns, but at a much higher level. In BASE 5×5 , cumulative contributions decrease from 67.8 in Round I to 52.1 in Round V ($p = 0.031$; Appendix C, Table C2 – Model 2). Again, as predicted, with ratcheting, the contributions in Round I of RAT 5×5 are significantly lower than those in BASE 5×5 (47.1 vs. 67.8, $p = 0.003$; Appendix C, Table C2 – Model 2). Even though the cumulative contributions in RAT 5×5 increase to 68.1 in Round V, which clearly exceeds the provision level of 52.1 in BASE 5×5 ($p = 0.037$; Appendix C, Table C2 – Model 2), the difference in average cumulative contributions between BASE 5×5 and RAT 5×5 is not statistically significantly different from zero (59.5 vs. 58.4, $p = 0.877$; Appendix C, Table C1).

Participants cooperate more when they can make many small interactive contribution decisions, rather than a few large ones. Fig. 4a shows that contributions to the public good start relatively low in the first decision of BASE 5×5 . Compared to the contributions in the first round in BASE 5×1 , participants contribute significantly less in the first decision of an average round in BASE 5×5 (46.4 vs. 17.7, $p < 0.001$; Appendix C, Table C3 – Model 1). However, the decline in contributions across decisions in the 5×5 -setting is much less pronounced than the decline in contributions across rounds in the 5×1 -setting ($p < 0.001$; Appendix C, Table C3 – Model 2). While contributions in BASE 5×1 decrease from 46.4 in Round I to 11.1 in Round V, contributions in BASE 5×5 decrease from 17.7 in Decision 1 to 7.7 in Decision 5. As a result, the cumulative contributions in the last decision of an average round in BASE 5×5 clearly exceed the average contributions per round in BASE 5×1 (59.5 vs. 31.5, $p < 0.001$; Appendix C, Table C1).

To ease interpretation of the results under the ratchet-up mechanism, we decompose the contributions in our ratcheting treatments into the required minimum contributions, which are equal to the contributions in the previous interaction, and the excess contributions, i.e., the contributions that exceed the minimum contributions. Fig. 4b shows that the excess contributions in the first decision of an average round in RAT 5×5 are substantially lower than the excess contributions in the first round of RAT 5×1 (3.1 vs. 20.9, $p < 0.001$; Appendix C, Table C3 – Model 3). While the excess contributions decline sharply over rounds in RAT 5×1 , they decline significantly less over decisions in RAT 5×5 ($p < 0.001$; Appendix C, Table C3 – Model 4). Consequently, the cumulative excess contributions in Decision 5 of an average round in RAT 5×5 are significantly higher than the average excess contributions in RAT 5×1 (13.6 vs. 7.1, $p < 0.001$; Appendix C, Table C3 – Model 6). This explains that the cumulative contributions at the end of an average round in RAT 5×5 are significantly higher than the average contributions in RAT 5×1 (59.5 vs. 28.7, $p < 0.001$; Appendix C, Table C1).

¹¹ In Appendix B and D (Fig. D5 for the 5×1 -setting and Fig. D6 for the 5×5 -setting), we discuss the distribution of our simulation results for the 1000 groups in greater detail. The median, for example, reflects the general trend in the average contributions in the 5×1 - and 5×5 -setting, both in BASE and RAT.

¹² Even though our simulations suggest directional effects, the analyses of the experimental data rely solely on two-sided tests.

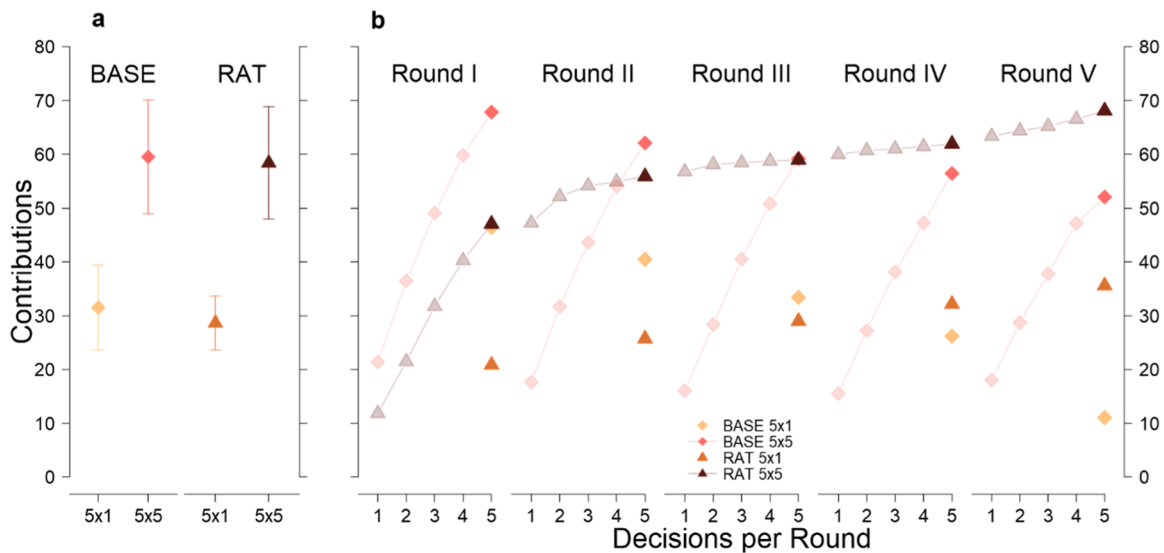


Fig. 3. Contributions to the public good.

Note: All results are based on $n = 368$ subjects participating in groups of four and randomly assigned to treatments: $n_{BASE\ 5 \times 1} = 88$, $n_{BASE\ 5 \times 5} = 96$, $n_{RAT\ 5 \times 1} = 88$, and $n_{RAT\ 5 \times 5} = 96$. Participants receive an endowment of $w = 100$ ECUs per round in all treatments. We use groups' average contribution to the public good, measured in ECU, as unit of observation. a, overall contributions by contribution mechanism (BASE vs. RAT) and setting (5×1 vs. 5×5), measured by the average of groups' contributions from all five rounds of the experiment. We report 90% confidence intervals based on robust estimates for the standard errors (Appendix C, Table C1). b, in 5×1 -settings, groups' contributions per round by contribution mechanism. In 5×5 -settings, groups' cumulative contributions per decision and round by contribution mechanism.

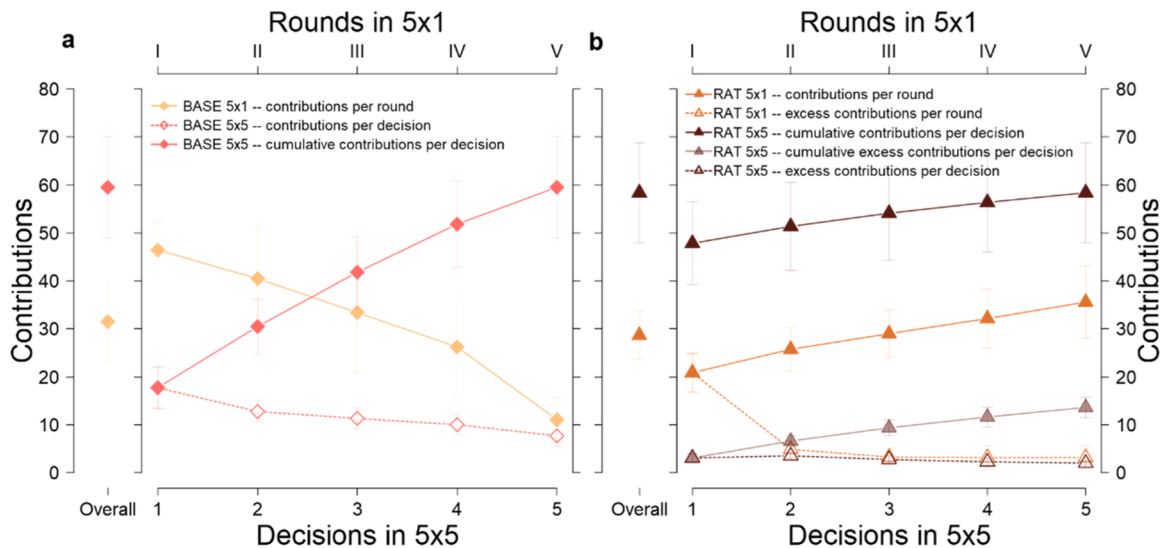


Fig. 4. Contributions to the public good per round and decision.

Note: a, we use groups' contributions in ECU per round and decision as unit of observation in BASE 5×1 and BASE 5×5 , respectively. In BASE 5×1 , we use groups' contributions per round. In BASE 5×5 , we use the average of groups' contributions in a given decision from all five rounds of the experiment. Moreover, we differentiate between the actual and the cumulative contributions per decision. We report 90% confidence intervals based on robust estimates for the standard errors (Appendix C, Table C3 – Model 1). b, we use groups' contributions per round as unit of observation in RAT 5×1 . In addition, we show the excess contributions, i.e., groups' contributions above the required minimum. In RAT 5×5 , we use the average of groups' cumulative contributions in a given decision from all five rounds as unit of observation. We also disentangle the cumulative contributions into the excess contributions and the cumulative excess contributions per decision. We report 90% confidence intervals based on robust estimates for the standard errors (Appendix C, Table C3 – Model 2).

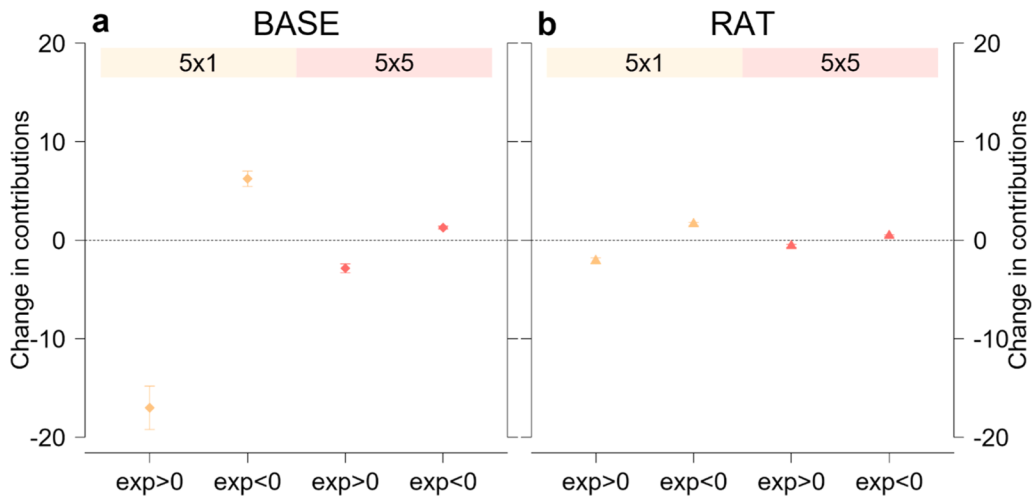


Fig. 5. Total effect of exploitation by mechanism and setting.

Note: **a**, shows the total effect of exploitation on participants' contributions to the public good in BASE 5×1 and BASE 5×5 . **b**, shows the total effect of exploitation on participants' excess contributions in RAT 5×1 and RAT 5×5 . All contributions are measured in ECU. The total effect of exploitation is given by the product of the level of exploitation (see Appendix D, Fig. D7) and the marginal effect of exploitation (see Appendix C, Table C4). Across the contribution mechanisms and settings, we separately report the effects for participants who have been free ridden by their group members ($\text{exp} > 0$) and those who have been free riding on their group members' contributions ($\text{exp} < 0$).

The reason why the level of contributions to the public good in the 5×5 -setting exceeds that in the 5×1 -setting, both in BASE and RAT, is shown in Fig. 4: Contributions start relatively low in the first decision of the 5×5 -settings, but they hardly decline from decision to decision. Thus, the cumulative contributions at the end of an average round in the 5×5 -settings are higher than the average contributions in the corresponding 5×1 -settings. This pattern is consistent with the interpretation that more frequent interactions allow subjects to build trust more safely over time. At the same time, we cannot fully exclude an alternative explanation based on salience or experimenter-demand effects (e.g., Zizzo, 2010): asking participants to make five contribution decisions per round may itself make cooperative behavior more focal. Our design does not separately identify these channels. Still, our results do not fit a simple monotone demand account. The 5×5 -settings' advantage arises because contributions decline much less across decisions than across rounds and because realized exploitation is substantially lower in the 5×5 -settings. We therefore view safer reciprocal adjustment as the main mechanism behind the difference between the 5×5 - and 5×1 -settings, while acknowledging that repeated prompting may also play a role.

Indeed, one measure of successful trust is the degree of exploitation, measured as the difference between participants' contributions to the public good and the average of the other group members' contributions. While a level of exploitation greater than zero implies that the participant has been free ridden by her group members, a level of exploitation less than zero means that she was free riding on her group members' contributions. Appendix D, Fig. D7 shows that in the 5×1 -settings participants have more reason to be concerned about exploitation than in the 5×5 -settings. The level of exploitation in the 5×1 -settings is much greater than in the 5×5 -settings. In BASE, the interquartile range is 35 in the 5×1 -setting and 5.83 in the 5×5 -setting. In RAT, the range between the 3rd and the 1st quartile is 30.33 and 3.58 in the 5×1 - and 5×5 -setting, respectively. In this sense, it is safer to cooperate in 5×5 -settings. One further result, summarized in Fig. 5, reveals important heterogeneities in the effect of exploitation: We find that the drop in contributions after being exploited clearly exceeds the lift in contributions after exploiting the other group members. To estimate the total effect of exploitation, we combine the level of exploitation with the marginal effect of exploitation. In this way, we create a uniform measure that combines the average level of exploitation with the marginal effect of exploitation.

Consistent with the empirical evidence that many subjects are conditional cooperators who are willing to cooperate if others cooperate as well (Fischbacher et al., 2001; Fischbacher and Gächter, 2010), we find that participants' contributions in a given interaction depend on their group members' contributions in the previous interaction of the experiment. More precisely, and in line with the literature on learning in social dilemma situations (Chaudhuri et al., 2017; Neugebauer et al., 2009), participants who have been free ridden by their other group members in a given interaction reduce their contributions to the public good in the subsequent interaction. In contrast, those participants who have been free riding on their group members' contributions, follow their group members and contribute more in the subsequent interaction. Most importantly, Fig. 5 shows two important heterogeneities: First, we find that the drop in contributions after being exploited clearly exceeds the lift in contributions after exploiting the other group members. For example, in BASE 5×1 , participants who have been free ridden by their group members reduce their contributions in the subsequent round by 17.2. In contrast, those participants who have been free riding on their group members' contributions,

increase their contributions by on average 6.2.¹³ This is in line with the literature showing that participants suffer more from disadvantageous than advantageous inequality (Blanco et al., 2011; Nunnari and Pozzi, 2022). Second, the effects are more pronounced in BASE than in RAT. To further explore the effects of exploitation, we estimate the total effects of exploitation separately for participants with a high (see Appendix D, Fig. D8) and low (see Appendix D, Fig. D9) level of trust.¹⁴ While the general pattern persists for all participants, the results show that the total effect of exploitation is especially pronounced for high-trust participants. In particular, high-trust participants respond more strongly to being free ridden than low-trust participants. In addition, since the level of exploitation is lower in the 5×5 -settings than in the 5×1 -settings, the decline in contributions from round to round in the 5×5 -settings is also smaller than the decline in contributions from round to round in the 5×1 -settings. Thus, the cumulative contributions in the 5×5 -settings are higher than the average contributions in the 5×1 -settings.

5. Conclusions

Addressing climate change is a collective challenge that hinges on cooperation. Trust is the key to such cooperation, and it largely relies on reciprocity (Fehr and Gächter, 2000; MacKay et al., 2015). Thus, embedding reciprocity within climate negotiations is critical (Cramton et al., 2017). In this context, we investigate two mechanisms underpinning the Paris Agreement - ratcheting and increased frequency of commitments. Our question cannot be answered by field studies. Instead, we employ simulations anchored in established patterns of (conditional) cooperative behavior to guide our thinking and to build hypotheses, and corroborate these with laboratory experiments.

Our findings reveal that the ratchet-up mechanism, perhaps contrary to expectations, does not promote cooperation in our setting. Instead, the mechanism heightens the risk of exploitation, deterring trust and cooperation. Conversely, increasing the frequency of commitments promotes cooperation. This design feature diminishes the fear of exploitation, allowing trust to emerge. The fundamental reason is that, as hypothesized by Elinor Ostrom and her colleagues in Poteete et al. (2010, p. 227), “[...] *trust and reciprocity are mutually reinforcing. [...] a decrease in either can generate a downward cascade leading to little or no cooperation [...]*”. More frequent interactions allow trust to evolve more safely, which in turn, escalates ambition and cooperation. Our analysis thus suggests that the recent intended policy shift towards more frequent interactions in the context of the Paris climate agreement is a strategic advance.

There are several ways to increase the frequency of interactions in climate negotiations, and recent developments suggest this direction is gaining momentum. For example, the current five-year NDC cycle under the Paris Agreement could include commitment reviews every two to three years. During these reviews, countries could strengthen their commitments. In fact, recognizing the need for reform, climate leaders have recently called for making COP meetings “*smaller, more frequent and solutions-driven*” to improve implementation and accelerate the pace of change (World Economic Forum, 2025). Additionally, smaller “*climate clubs*” or bilateral partnerships with more frequent negotiation schedules could serve as laboratories for testing increased ambition before scaling up to broader multilateral forums (Nordhaus, 2015; Wolfram et al., 2025). The G7 Climate Club, launched in December 2022 and now comprising 46 member states as of May 2025, exemplifies this approach by focusing on industrial decarbonization and cooperative climate action outside the traditional UNFCCC framework. This may also include sectoral agreements (e.g., for aviation, shipping, or heavy industry) that would allow experimentation with higher-frequency interactions within more homogeneous groups, which – as our research suggests – could reduce vulnerability to exploitation and allow cooperation to escalate more safely over time. Further research should explore additional behavioral mechanisms that could reinforce trust and reciprocity. Changing the framing of climate negotiations from a distributive carbon budget to a cooperation framework could mitigate the adversarial dynamics of zero-sum scenarios. Moreover, the empirical literature suggests that reciprocal cooperation is unlikely to robustly arise within large, heterogeneous groups. A phased strategy, perhaps initiating with a “*climate club*” (Nordhaus, 2015) or strategic bilateral partnerships, may be an effective complement in fostering an environment of safe trust and reciprocity, not much unlike a strategy of more frequent interactions can foster such an environment. Reciprocity may also be strengthened by a shared understanding of mutual expectations. While such an understanding may not be necessary in symmetric laboratory dilemma games, where norms of fairness occur naturally, explicit agreements on contributions in asymmetric environments may be useful to crystallize these expectations, guiding complex negotiations towards a common goal. Moreover, the question of whether an explicit reversal of commitments is possible and how it affects international climate cooperation may also become increasingly important beyond our laboratory setting. Further experiments could also test the robustness of our results to even more interactions and focus on the channels through which more frequent interactions foster cooperation. For example, there is no clear evidence that more frequent interactions are only effective through contributions (Duffy et al., 2007). More frequent contributions could affect cooperation per se or through the additional information and feedback that participants receive about their group members. Our design does not allow us to further decompose these different channels. We leave such questions to further research.

¹³ We find that both the marginal (Appendix C, Table C4) and total effects of exploitation (Appendix C, Table C5) are robust to controlling for participants' demographic and socio-economic characteristics, including age, gender, income, and field of study.

¹⁴ We elicited participants' level of trust on a scale from 0 (low) to 10 (high) in the ex-post questionnaire to our experiment and classify participants with a reported level of trust above the median (>6) as high-trust and those with a trust level at or below the median (≤ 6) as low-trust participants. The Supplementary Material provides the exact wording of the question.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jebo.2026.107559](https://doi.org/10.1016/j.jebo.2026.107559).

Data availability

Data will be made available on request.

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